Guideline to make and understand Unit Test Case

1. Overview

- In the template, Unit test cases are based on functions. Each sheet presents test cases for one function.
- Cover: General information of the project and Unit Test cases
- FunctionList: The list of Classes and Functions in the document.
  + To control that the number of Unit TC meets customer's requirement or the norm, user should fill value for 'Normal number of Test cases/KLOC'.
  + Click on Function link to open the related Test cases of the function.
  Note: You should create new Function sheet before creating the link
- Test Report: provive the overview results of Functions Unit test: Test coverage, Test successful coverage (Summary, for normal/abnormal/boundary cases)

Note: Should check the formula of "Sub Total" if you add more functions

2. Content in Test function sheet

2.1 Combination of test cases.

- To verify that number of Unit TC meets customer's requirement or not. User has to fill number LOC of tested function and fill value of 'Normal number test cases/KLOC' FunctionList sheet, which is required by customer or normal value. The number of lacked TC is shown in 'Lack of test cases' item.
- If the number of Unit TC does not meet the requirement, creator should explain the reasons.
- If the number of 'Normal number test cases/KLOC' item in FunctionList sheet is not recorded, the number in 'Lack of test cases' is not calculated.

2.2 Condition and confirmation of Test cases.

Each test case is the combination of condition and confirmation.

a. Condition:

- Condition is combination of precondition and values of inputs.
- Precondition: it is setting condition that must exist before execution of the test case.
  Example: file A is precondition for the test case that needs to access file A.
- Values of inputs: it includes 3 types of values: normal, boundary and abnormal.
  . Normal values are values of inputs used mainly and usually to ensure the function works.
  . Boundary values are limited values that contain upper and lower values.
  . Abnormal values are non-expected values. And normally it processes exception cases.
- For examples:
  Input value belongs to 5<= input <=10.
  . 6,7,8,9 are normal values.
  . 5, 10 are boundary values.

. -1, 11,... are abnormal values.

b. Confirmation:

- It is combination of expected result to check output of each function.
  If the results are the same with confirmation, the test case is passed, other case it is failed.
- Confirmation can include:
  + Output result of the function.
  + Output log messages in log file.
  + Output screen message...

c. Type of test cases and result:

- Type of test case: It includes normal, boundary and abnormal test cases. User selects the type based on the type of input data.
- Test case result: the actual output results comparing with the Confirmation.
  P for Passed and F for Failed cases.
  It can 'OK' or 'NG' (it depends on habit of the teams or customers)

2.3. Other items:

- Function Code: it is ID of the function and updated automatically according to FunctionList sheet.
- Function Name: it is name of the function and updated automatically according to FunctionList sheet.
- Created By: Name of creator.
- Executed By: Name of person who executes the unit test
- Lines of code: Number of Code line of the function.
- Test requirement: Brief description about requirements which are tested in this function, it is not mandatory.

UNIT TEST DOCUMENT

Project Name	*Develop an AI-Assisted Omnichannel Order Fulf*	**Creator**	Nguyễn Phương Mỹ Thuận
Project Code	*FA26SE036*	**Issue Date**	*25/10/2026*
Document Code	*FA26SE036_Unit Test_v1.0*	**Version**	*1.0*

Record of change

Effective Date	Version	Change Item	*A,D,M	Change description	Reference
25/10/2026	1.0	Whole document	A	Unit test cases of 5 core back-end functions, mapped to	*Report 3 (SRS), Report 4 (SDD), Report5_Test_Report.xlsx*

Function List

Project Name	*Develop an AI-Assisted Omnichannel Order Fulfillment and Packaging Optimization System (OptiPack AI)*
Project Code	*FA26SE036*
Normal number of Test cases/KLOC	*100*
Test Environment Setup Description	*1. Node.js 22, Jest 30 + ts-jest (npm run test in be/)*

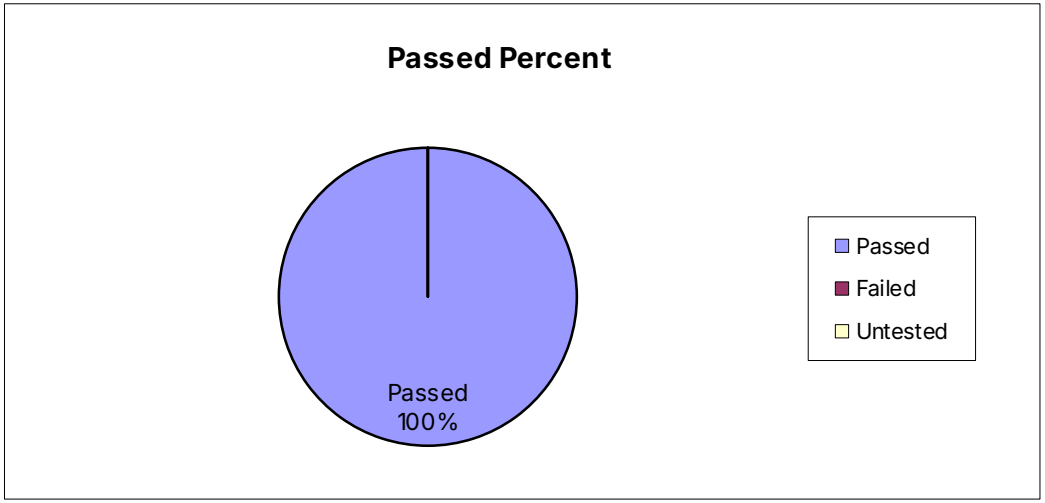
No	Requirement Name	Class Name	Function Name	Function Code(Optional)	Sheet Name	Description	Pre-Condition
1	Staff login with e-mail / password, lockout, 72-hour tempora	AuthService	login	UT-AUTH-LOGIN	UT-AUTH-LOGIN	Staff login with e-mail / password, lockout, 72-hour temporary password, optional two-factor (TOTP, backup code, trusted device) – SRS 3.2.1	Can connect to MongoDB (mocked model)
2	Scan one item while picking: SKU of the group, ordered quant	OrderGroups Service	pickItem	UT-PICK-ITEM	UT-PICK-ITEM	Scan one item while picking: SKU of the group, ordered quantity, FIFO lot, idempotent scan – SRS 3.5.1	MongoDB transaction available (mocked)
3	Allowed Order Group status transitions (33) and rejection of	allowed-status-transitions	isValidStatusTransition / getAllowedNextStatuses	UT-GROUP-TRANS	UT-GROUP-TRAN	Allowed Order Group status transitions (33) and rejection of invalid jumps – SDD 3.8	Pure function, no external dependency
4	Return ticket rules: 15-day window, one open ticket, no self	ReturnsService	createSimulated / approve / reject / receive / inspect	UT-RETURNS	UT-RETURNS	Return ticket rules: 15-day window, one open ticket, no self-approval, reason required, inspection – SRS 3.8, BR-11, BR-12	MongoDB transaction available (mocked)
5	3D packing plan per order: valid placement, determinism, fra	solve-order (BRKGA + EMS)	solveOrder	UT-SOLVER	UT-SOLVER	3D packing plan per order: valid placement, determinism, fragile items, folding, box stock and exclusions – SRS 3.6.1, BR-06, BR-15	Box catalog with stock (fixture)

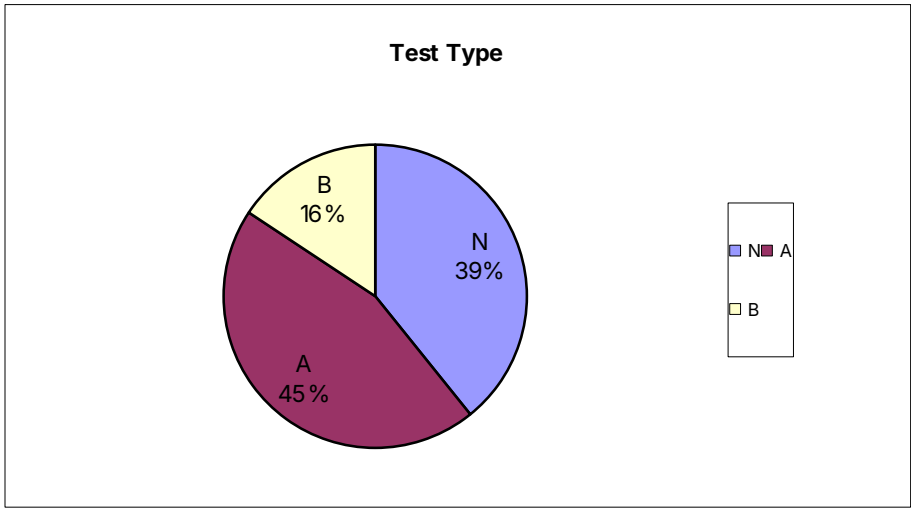
UNIT TEST REPORT

Project Name	*Develop an AI-Assisted Omnichannel Order Fulfillment an*	**Creator**	**Nguyễn Phương Mỹ Thuận**
Project Code	*FA26SE036*	**Reviewer/Approver**	**Thân Thị Ngọc Vân**
Document Code	*FA26SE036_Unit Test Report_v1.0*	**Issue Date**	*25/10/2026*
Notes	*Review 2 release (tag review-2): unit tests of 5 core back-end functions; all cases run automatically in CI (Jest).*		

No	Function code	Passed	Failed	Untested	N	A	B	Total Test Cases
1	UT-AUTH-LOGIN	13	0	0	5	7	1	13
2	UT-PICK-ITEM	8	0	0	5	2	1	8
3	UT-GROUP-TRANS	8	0	0	4	2	2	8
4	UT-RETURNS	12	0	0	3	8	1	12
5	UT-SOLVER	10	0	0	3	4	3	10
	Sub total	51	0	0	20	23	8	51

Test coverage	**100.00** %
Test successful coverage	**100.00** %
Normal case	**39.22** %
Abnormal case	**45.10** %
Boundary case	**15.69** %





Function Code	UT-AUTH-LOGIN	Function Name	AuthService.login			
Created By	Thuận	Executed By	GitHub Actions CI (run #258, commit 1a3101c)			
Lines of code	143	Lack of test cases	1.3			
Test requirement	Staff login with e-mail / password, lockout, 72-hour temporary password, optional two-factor (TOTP, backup code, trusted device) – SRS 3.2.1. Test file: be/src/modules/auth/auth.service.spec.ts; source: be/src/modules/auth/auth.service.ts					
Passed	Failed	Untested	N/A/B			Total Test Cases
13	0	0	5	7	1	13

		UTCID01	UTCID02	UTCID03	UTCID04	UTCID05	UTCID06	UTCID07	UTCID08	UTCID09	UTCID10	UTCID11	UTCID12	UTCID13		
Condition	**Precondition**															
	Can connect to MongoDB (mocked model)	O	O	O	O	O	O	O	O	O	O	O	O	O		
	User															
	exists, active	O		O				O	O	O	O	O	O	O		
	does not exist		O													
	inactive					O										
	locked				O											
	temporary password > 72 h						O									
	Password															
	correct	O	O		O	O	O	O	O	O	O	O	O	O		
	wrong			O												
	Two-factor															
	disabled	O														
	enabled, no code							O								
	enabled, wrong TOTP								O							
	enabled, valid TOTP									O						
	enabled, MFA secret empty										O					
	enabled, trusted device											O				
	enabled, valid backup code												O			
	enabled, wrong backup code													O		
Confirm	**Return**															
	token pair	O								O		O	O			
	{ mfa_required: true }							O								
	new trusted_device_token									O						
	backup code removed												O			
	Exception															
	401 Unauthorized		O	O		O			O		O			O		
	403 AUTH_ACCOUNT_LOCKED				O											
	403 password deadline						O									
	Log message															
	Test name in auth.service.spec.ts (line)	line 115	line 140	line 153	line 170	line 181	line 197	line 217	line 235	line 254	line 276	line 295	line 317	line 338		
Result	Type(N : Normal, A : Abnormal, B : Boundary)	N	A	A	A	A	B	N	A	N	A	N	N	A		
	Passed/Failed	P	P	P	P	P	P	P	P	P	P	P	P	P		
	Executed Date	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026		

Function Code	*UT-PICK-ITEM*	Function Name	OrderGroupsService.pickItem
Created By	*Thi, Thuận*	Executed By	*GitHub Actions CI (run #258, commit 1a3101c)*
Lines of code	*207*	Lack of test cases	12.7
Test requirement	*Scan one item while picking: SKU of the group, ordered quantity, FIFO lot, idempotent scan – SRS 3.5.1. Test file: be/src/modules/order-groups/order-groups.service.pickItem.spec.ts; source: be/src/modules/order-groups/order-groups.service.ts*		

Passed	Failed	Untested	N/A/B			Total Test Cases
8	0	0	5	2	1	8

		UTCID01	UTCID02	UTCID03	UTCID04	UTCID05	UTCID06	UTCID07	UTCID08
Condition	**Precondition**								
	MongoDB transaction available (mocked)	O	O	O	O	O	O	O	O
	Group status								
	picking	O	O	O	O	O	O		O
	not picking							O	
	Scanned SKU								
	in group		O	O		O	O	O	O
	not in group	O							
	same SKU in another shop				O				
	Quantity								
	≤ ordered		O	O	O	O			
	> ordered						O		
	Bin								
	given		O						
	not given (oldest lot)					O			
	client_event_id								
	new	O	O	O	O	O	O	O	
	already processed								O
	Scan								
	first of the group			O					
	later scan		O						
Confirm	**Return**								
	stock decremented + pick_event		O	O	O	O			
	lot recorded on pick_event					O			
	website "preparing" event once			O					
	previous result returned								O
	Exception								
	ORD_GROUP_ITEM_NOT_IN_GROUP	O							
	ORD_GROUP_PICK_EXCEEDS_ORDERED						O		
	ORD_GROUP_PICK_NOT_ALLOWED							O	
	Log message								
	Test name in order-groups.service.pickItem.spec.ts (line)	**line 131**	**line 143**	**line 174**	**line 194**	**line 216**	**line 286**	**line 299**	**line 309**
Result	Type(N : Normal, A : Abnormal, B : Boundary)	A	N	N	N	N	B	A	N
	Passed/Failed	P	P	P	P	P	P	P	P
	Executed Date	**25/10/2026**	**25/10/2026**	**25/10/2026**	**25/10/2026**	**25/10/2026**	**25/10/2026**	**25/10/2026**	**25/10/2026**

Function Code	UT-GROUP-TRANS	Function Name	allowed-status-transitions.isValidStatusTransition / getAllowedNextStatuses
Created By	Thi, Thuận	Executed By	GitHub Actions CI (run #258, commit 1a3101c)
Lines of code	113	Lack of test cases	3.3
Test requirement	Allowed Order Group status transitions (33) and rejection of invalid jumps – SDD 3.8. Test file: be/src/modules/order-groups/enums/allowed-status-transitions.spec.ts; source: be/src/modules/order-groups/enums/allowed-status-transitions.ts		

Passed	Failed	Untested	N/A/B			Total Test Cases
8	0	0	4	2	2	8

		UTCID01	UTCID02	UTCID03	UTCID04	UTCID05	UTCID06	UTCID07	UTCID08
Condition	**Precondition**								
	Pure function, no external dependency	O	O	O	O	O	O	O	O
	From → to								
	happy path step (e.g. picking → picked)	O							
	pending_approval → picked		O						
	packed → canceled			O					
	shipped → packed				O				
	awaiting_packaging → packed					O			
	packed → packed						O		
	shipped → canceled							O	
	canceled → any								O
Confirm	**Return**								
	true	O	O	O	O				
	false					O	O	O	
	[] (no next status)								O
	Exception								
	Log message								
	Test name in allowed-status-transitions.spec.ts (line)	line 29	line 35	line 154	line 176	line 115	line 115	line 147	line 164
Result	Type(N : Normal, A : Abnormal, B : Boundary)	N	N	N	N	A	B	A	B
	Passed/Failed	P	P	P	P	P	P	P	P
	Executed Date	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026

Function Code	UT-RETURNS	Function Name	ReturnsService.createSimulated / approve / reject / receive / inspect
Created By	Thi, Thuận	Executed By	GitHub Actions CI (run #258, commit 1a3101c)
Lines of code	351	Lack of test cases	23.1
Test requirement	Return ticket rules: 15-day window, one open ticket, no self-approval, reason required, inspection – SRS 3.8, BR-11, BR-12. Test file: be/src/modules/shipments/returns.service.spec.ts; source: be/src/modules/shipments/returns.service.ts		

Passed	Failed	Untested	N/A/B			Total Test Cases
12	0	0	3	8	1	12

			UTCID01	UTCID02	UTCID03	UTCID04	UTCID05	UTCID06	UTCID07	UTCID08	UTCID09	UTCID10	UTCID11	UTCID12			
Condition	Precondition																
		MongoDB transaction available (mocked)	O	O	O	O	O	O	O	O	O	O	O	O			
	Order group																
		delivered ≤ 15 days				O	O	O									
		delivered > 15 days			O												
		not delivered		O													
		AURELLE (marketplace-managed)	O														
	Ticket																
		none open	O	O	O	O		O									
		open ticket exists					O										
		requested							O	O	O	O	O				
		approved (awaiting receipt)												O			
	Items																
		valid	O	O	O		O	O									
		more than bought / unknown SKU				O											
		inspected fewer than returned												O			
	Actor																
		creator							O								
		other user								O	O	O					
	Note																
		given															
		missing										O					
	Type																
		return & refund									O						
		refund only								O							
Confirm	Return																
		ticket "requested"						O									
		ticket closed								O							
		ticket awaiting_receipt									O						
	Exception																
		409 RMA_MARKETPLACE_MANAGED	O														
		409 RMA_ORDER_NOT_DELIVERED		O													
		409 RMA_WINDOW_EXPIRED			O												
		400 RMA_INVALID_ITEMS				O											
		409 RMA_OPEN_EXISTS					O										
		403 RMA_SELF_APPROVAL							O								
		400 RMA_NOTE_REQUIRED										O					
		409 RMA_INVALID_STATUS											O				
		400 RMA_INSPECTION_MISMATCH												O			
	Log message																
		Test name in returns.service.spec.ts (line)	line 62	line 68	line 73	line 78	line 101	line 106	line 113	line 118	line 118	line 129	line 134	line 157			
Result	Type(N : Normal, A : Abnormal, B : Boundary)		A	A	B	A	A	N	A	N	N	A	A	A			
	Passed/Failed		P	P	P	P	P	P	P	P	P	P	P	P			
	Executed Date		25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026			

Function Code	UT-SOLVER	Function Name	solve-order (BRKGA + EMS).solveOrder
Created By	Thi	Executed By	GitHub Actions CI (run #258, commit 1a3101c)
Lines of code	590	Lack of test cases	49
Test requirement	3D packing plan per order: valid placement, determinism, fragile items, folding, box stock and exclusions – SRS 3.6.1, BR-06, BR-15. Test file: be/src/modules/packing/solver/solver.spec.ts; source: be/src/modules/packing/solver/solve-order.ts		

Passed	Failed	Untested	N/A/B			Total Test Cases
10	0	0	3	4	3	10

		UTCID01	UTCID02	UTCID03	UTCID04	UTCID05	UTCID06	UTCID07	UTCID08	UTCID09	UTCID10
Condition	**Precondition**										
	Box catalog with stock (fixture)	O	O	O	O	O	O	O	O	O	O
	Order										
	random items (25 orders)	O									
	fixed small order		O					O	O	O	
	≤ 3 items			O							
	> 3 items				O						
	fragile item					O					
	jean 26 cm wide (box M 25 cm)						O				
	oversize / overweight item										O
	Run										
	same input twice		O								
	Box stock										
	all in stock	O	O	O	O	O	O				O
	best box out of stock							O			
	box excluded								O		
	all fitting boxes out of stock									O	
Confirm	**Return**										
	plan passes validator	O									
	identical coordinates		O								
	exhaustive search			O							
	BRKGA				O						
	nothing on top of fragile item					O					
	item folded to use box M						O				
	another box used							O			
	excluded box not used								O		
	Exception										
	OUT_OF_STOCK									O	
	ITEM_TOO_LARGE / ITEM_TOO_HEAVY										O
	Log message										
	Test name in solver.spec.ts (line)	line 41	line 58	line 67	line 67	line 79	line 110	line 125	line 133	line 138	line 146
Result	Type(N : Normal, A : Abnormal, B : Boundary)	N	N	B	B	N	B	A	A	A	A
	Passed/Failed	P	P	P	P	P	P	P	P	P	P
	Executed Date	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026	25/10/2026